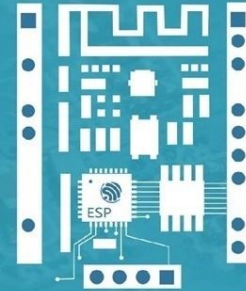


Get Started with Arduino IDE

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1 Download CH340 Driver

- Method ①:
Search, download driver. Recommended for Win10 OS



CH340 driver



Microsoft Bing

CH340 driver



- Method ②: CH340 driver official website to download:
<https://www.wch-ic.com/downloads/category/67.html>

Driver

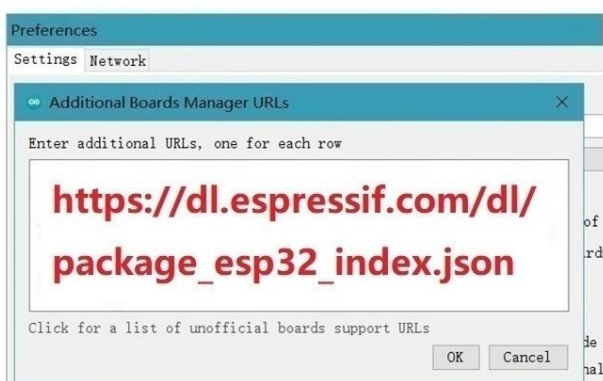
Function type: **USB to UARTs** Chip model: **CH340** **Q Search**

Downloads (5)

Resource name	Resource Brief
CH341SER.EXE	CH340/CH341 USB to serial port One-Key installation VCP vendor driver for Windows, supports Windows 11/10/8.1/8/7/VISTA/XP/2000/98, Server 2025/2022/2019/2016/2012/2008/2003 -32/64bit, Microsoft WHQL/HCK Certified, supports USB to 3-line and 9-line serial port. Used to distribute to user with the product.
CH341SER.ZIP	CH340/CH341 USB to serial port Windows driver, supports Windows 11/10/8.1/8/7/VISTA/XP
CH341SER_ANDROID.ZIP	USB to Serial ports Android driver-free install, lib and demo application, supports CH340/ CH341/ CH342/ CH343/ CH344/ CH346/ CH347/ CH9101/ CH9102/ CH9103/ CH9104/ CH9111/ CH9114/ CH9143, for Android OS 4.4 and above in USB Host mode, without loading Android kernel driver and without root access operation. Includes apk installer, lib library file (Java Driver), App Demo(USB to UART Demo SDK).
CH341SER_LINUX.ZIP	Linux driver for USB to serial port, supports CH340 and CH341, supports 32/64-bit operating systems.
CH341SER_MAC.ZIP	USB to Serial ports macOS driver, supports CH340/ CH341/ CH342/ CH343/ CH344/ CH346/ CH347/ CH9101/ CH9102/ CH9103/ CH9104/ CH9111/ CH9114/ CH9143/ CH9149, supports OS X 10.9~10.15, OS X 11(Big Sur) and above, include installation guide document.

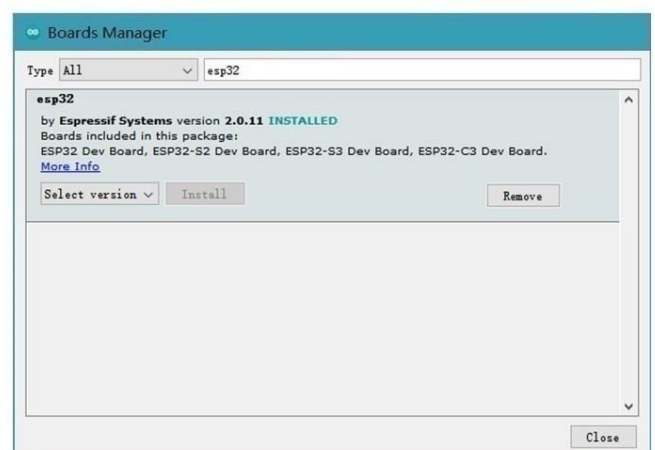
2 Add Boards Manager URLs in IDE

- IDE -> "Preferences" -> "Additional Boards Manager URLs"
Copy the link in red to your IDE



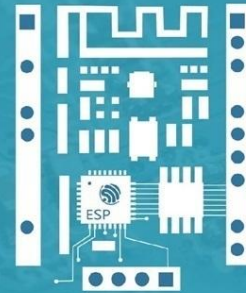
3 Install ESP32 Board Package

- IDE -> "Board" -> "Boards Manager"



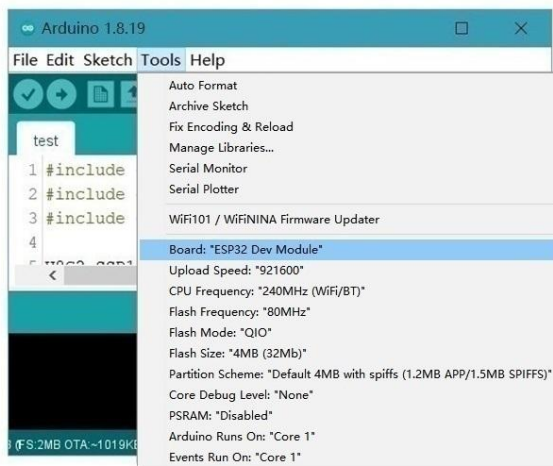
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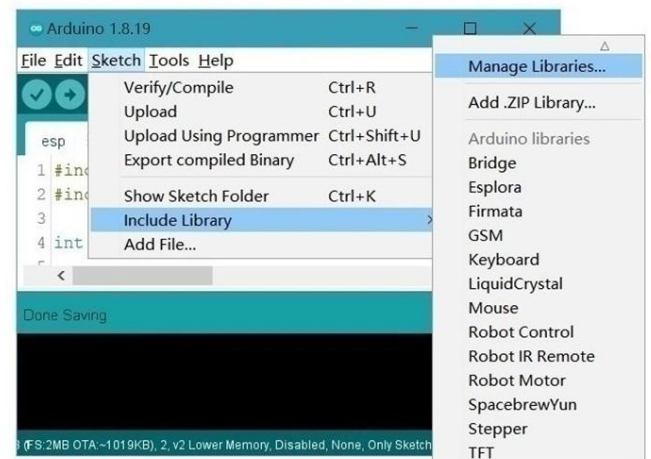
4 Select ESP32 Board

• IDE -> "Tools" -> "Board: ESP32 Dev Module"



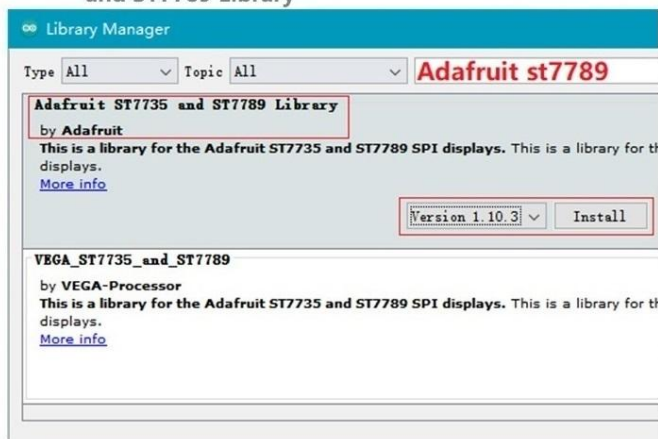
5 Include Library

• IDE -> "Sketch" -> "Include Library" -> "Manage Libraries"



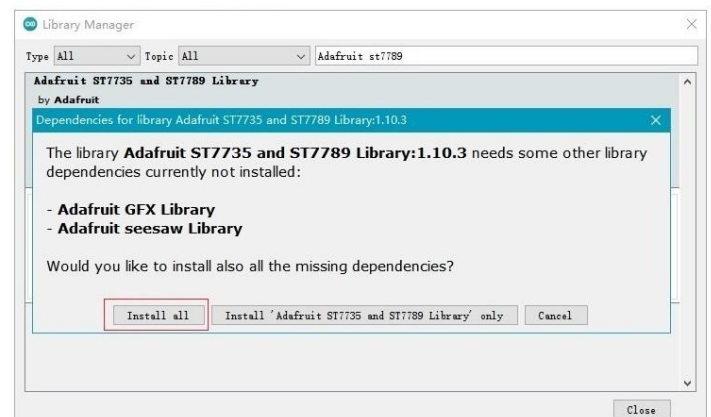
6 Install "ST7789" Library

• Search "Adafruit st7789", install "Adafruit ST7735 and ST7789 Library"



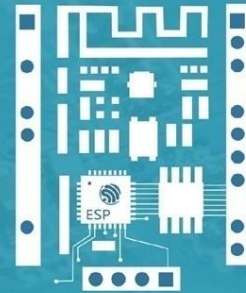
7 Install "ST7789" Library

• Install all libraries related to "Adafruit ST7735 and ST7789 Library"



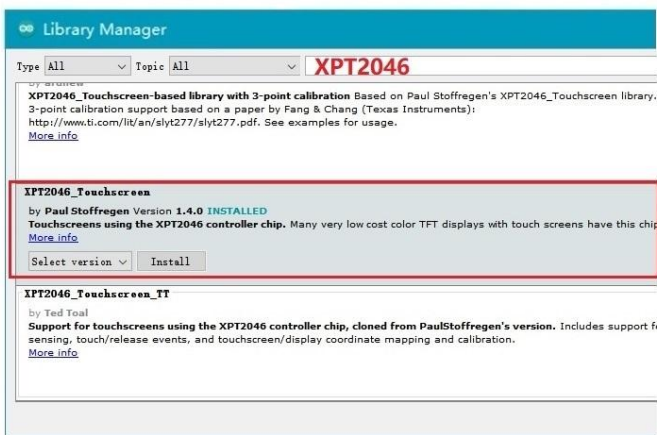
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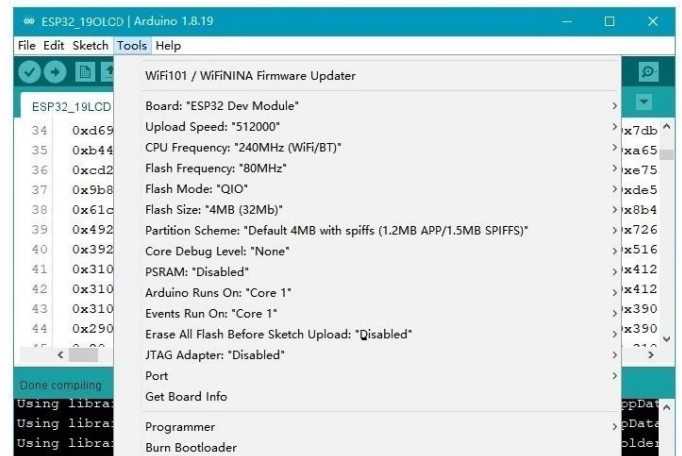
8 Install "XPT2046" Library

- Search "XPT2046", install "XPT2046_TouchScreen" by Paul Stoffregen



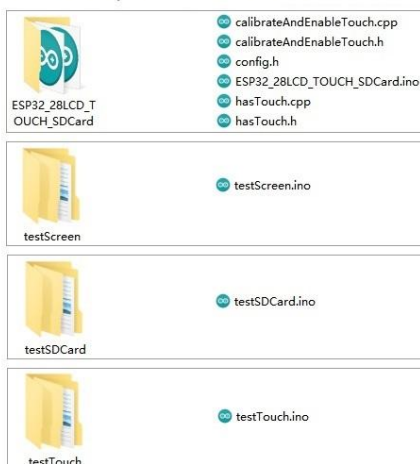
9 Set Board and Processor

- Configure the Tools parameters as shown in the figure



10 Sample Code

- We offer 4 types of sample code. Import the .ino files from the folder and upload the code to the board



11 Screen Color Setting

- Two panel types are used on this board, Check the code settings to make the screen display normally

- Open `ESP32_28LCD_TOUCH_SDCard.ino` / `testScreen.ino` and find this line near the top:

```
bool isDisplayInverted = true;
```

- It controls `lcdScreen.invertDisplay()`. Upload the code and look at the screen.

- Colors correct → keep `true`. Colors look like a photo negative → change it to `false` and upload again.